



CU Boulder - Data Science

Capstone Project Description: Q2 Impact



Company Name: Q2 Impact, LLC

Company Address: 2311 Wilson Blvd, 3rd Fl., Arlington, VA 22201

Industry Description and Company Products: Q2 Impact operates within the international development and consulting industry, specializing in delivering data-driven solutions to address complex global challenges. With over 25 years of experience, the company has provided insightful services across nearly 100 countries, focusing on areas such as monitoring and evaluation, global health, and learning.

The company's suite of services includes:

- **Monitoring and Evaluation:** Q2 Impact assists clients in collecting, analyzing, and utilizing data to assess program effectiveness and inform decision-making processes.
- **Global Health Initiatives:** The firm offers expertise in health program implementation and evaluation, aiming to improve health outcomes in diverse populations.
- **Learning and Capacity Building:** Q2 Impact provides training and development programs to enhance the skills and knowledge of organizations and individuals, fostering sustainable development.

By leveraging its extensive experience and specialized services, Q2 Impact contributes to transforming lives through knowledge-based solutions, addressing critical issues in international development.

Primary Contact(s)/Mentor(s) Names and Emails:

1. Kirsten Pecua (she/her): kpecua@q2impact.com, kirsten.n.pecua.ctr@mail.com
2. Marsha Grodnicki (she/her): mgrodnicki@q2impact.com

Planned Meeting Schedule with Students

Meeting Venue: Microsoft Teams

Meeting Occurrence Rate: Biweekly

Attention Measure: Moderate contact with the team. Once tasks and data sources are clarified, I expect the team to work autonomously unless assistance is needed outside of the biweekly discussions.

Requested Team Size: Two teams comprised of 3-5 students each.

Specific Required/Expected Deliverables: Geospatial data visualization platform.

Required Intellectual Property: None.

Project Introduction: The WPS (Women, Peace, and Security) team at NORTHCOM faces significant challenges in efficiently obtaining, sharing, and utilizing data about women's health and reproductive health services during humanitarian crises within its Area of Responsibility (AOR), which includes the United States, Canada, Mexico, and the Bahamas. One of the primary obstacles is the lack of up-to-date, localized information on resources (e.g. women's health centers and women-only shelters), which are critical for addressing the needs of women in times of crisis. First responders—such as the National Guard, FEMA, or local emergency teams—often struggle to quickly access information about the availability of services like maternal health clinics, reproductive health services, and specialized healthcare for women. The ability to quickly locate and verify the status of such resources, including their capacity, availability, and operational hours, is critical to providing an effective emergency response. To address this gap, we aim to develop a geospatial data visualization platform focused on resources for women. This tool will create an interactive, real-time map/dashboard that shows where these critical services are located, how to contact them, and whether they are operational and available during a crisis.

List of Project Functionalities (Functional Elements – what will be created, built, examined, discovered, predicted, etc.)

1. Data Scraping and Verification:

- Automated processes to gather and verify data from national and local health databases, and other open-source datasets related to resources for women in need during times of crises.
- Data will include locations, contact details, capacity, and real-time operational status (open, closed, capacity, hours, etc.).

2. Interactive Map:

- A user-friendly, geospatial map that visualizes the locations of women's health centers and shelters (e.g., maternal health clinics, family planning centers, prenatal care, and emergency reproductive health services).

3. Resource Details:

- Each resource will include detailed information on:

- **Type of service:** Reproductive health, maternal health, prenatal care, family planning, etc.
 - **Contact details:** Phone numbers, website links, and other contact information.
 - **Operational status:** Whether the center is open or closed, its working hours, and any temporary closures due to a crisis.
4. **Collaboration and Data Sharing:**
- The platform will support real-time data sharing between first responders, local agencies, and public health organizations.
 - Multiple stakeholders (National Guard, FEMA, local health agencies) will have the ability to update and share data during a crisis, ensuring that the map reflects the most current information on reproductive health services.
5. **User Interface:**
- A simple, intuitive interface for emergency responders and healthcare providers to quickly access, update, and utilize information.
 - Filtering and search functionalities to easily find nearby resources based on service type (e.g., maternal health, reproductive care) and availability.

List Dataset Needs and Availability

Health and Reproductive Services Data:

- Locations of women's health centers, maternal health clinics, and women-only shelters.
- Contact details, including phone numbers, websites, and addresses.
- Operational status (open/closed, hours of operation, and temporary closures during crises).
- Service capacity and available resources (e.g., number of beds, staffing levels).
- Types of services offered (e.g., prenatal care, reproductive health, family planning).

Geospatial Data:

- Mapping coordinates for health service locations.
- Localized demographic data to contextualize service coverage and needs.
- Real-time updates on accessibility (e.g., road conditions, crisis impact).

Crisis Management Data:

Data from emergency response agencies (e.g., FEMA, National Guard) on available resources.

Real-time updates on the status of healthcare facilities during disasters.

Dataset Availability:

Existing Data Sources:

- **National Health Databases:** Open data from agencies like HHS (Health and Human Services), CDC, or NIH that catalog health facilities.
- **State and Local Health Department Data:** Publicly available datasets on women's health services by state or region.
- **NGO Resources:** Data from organizations like Planned Parenthood or local women's shelters.

- **Global Databases:** WHO or UN data on reproductive health services for crisis-affected areas.
- **Geospatial Data:** OpenStreetMap or other GIS datasets for base mapping.

Data Students May Need to Gather:

- Real-time operational status from individual facilities via web scraping, APIs, or outreach.
- Updates from local emergency response teams during crises for dynamic status changes.
- Missing geospatial data, which may require manual input or integration.

Data Privacy Issues:

- All data for this project will be sourced from open-source platforms and publicly available databases.
- There are no anticipated data privacy issues, as no sensitive or personally identifiable information (PII) will be collected or utilized.
- Open-source tools and APIs ensure compliance with data use and sharing policies.

Project Notes, Requirements, Expectations

- **Project Background and Objective: Company Overview:**
Q2 Impact is a global consulting firm specializing in data-driven solutions for international development. With a focus on monitoring and evaluation, global health, and capacity building, Q2 Impact supports partners in addressing complex global challenges through innovative approaches.
- **Project Overview:**
This project aims to address a critical challenge faced by the WPS (Women, Peace, and Security) team at NORTHCOM: the lack of real-time, localized data on women's health and reproductive services during humanitarian crises. The solution will involve developing a geospatial data visualization platform that allows first responders to quickly access and update information on women's health services.
- **Objective:**
The primary objective is to create an interactive, real-time mapping and dashboard tool to support emergency responders in locating and verifying operational women's health resources. This platform will enhance the efficiency of disaster response efforts and improve outcomes for affected women and communities.

Project Requirements:

- Develop a geospatial platform to visualize real-time data on women's health resources (e.g., clinics, shelters, and health services).
- Implement automated processes for scraping, verifying, and updating resource information.

- Design a user-friendly interface with filtering options to allow emergency responders to locate specific services.
- Ensure open-source data integration to avoid privacy issues and facilitate updates by multiple stakeholders.
- Collaborate with emergency response teams to refine the platform's functionality and usability.
- Constraints:
- Platform development must utilize only open-source tools and publicly available datasets.
- The solution must operate in real-time and provide accurate, reliable data during crises.

Why students should choose this project:

- **Real-World Impact:** Students will work on a project with tangible benefits, directly supporting emergency response operations and improving access to critical healthcare services for women during crises.
- **Interdisciplinary Skills:** The project integrates geospatial data science, software development, and user interface design, offering students opportunities to apply and expand their technical skills.
- **High-Value Experience:** Working with Q2 Impact and NORTHCOM offers exposure to cutting-edge humanitarian solutions and collaboration with industry leaders.
- **Innovation-Driven:** The project allows creative problem-solving through the development of a platform that addresses a unique and pressing need in humanitarian response.

Deliverables:

- **Geospatial Platform:** An interactive, real-time mapping tool displaying women's health resources.
- **User Interface:** A functional and intuitive interface with filtering options for users to locate specific services.
- **Presentation:** A final presentation showcasing the platform's capabilities, use cases, and implementation potential.

Skill Requirements:

- **Geospatial Data Science:** Experience with GIS tools like QGIS, OpenStreetMap, or Google Maps APIs.
- **Software Development:** Knowledge of programming languages such as Python, JavaScript, or R for data scraping and visualization.

- **Database Management:** Familiarity with SQL or NoSQL databases for storing and managing resource information.
- **User Interface Design:** Skills in developing user-friendly interfaces with a focus on accessibility and usability.
- **Collaboration:** Ability to work effectively in a team and communicate with external stakeholders to ensure project alignment.

Images: Images included in e-mail.

Other Questions to Consider and Note:

1) What is the nature of the project?

This project is **goal-oriented with specific development objectives** focused on creating a geospatial data visualization platform to support real-time decision-making in humanitarian crises. Students will work to design and implement a tool that collects, verifies, and visualizes critical data on women's health and reproductive services. The project will involve development tasks such as data scraping, database integration, geospatial mapping, and user interface design. While the end goals are well-defined, students will have flexibility in the methods and technologies they use to achieve them.

2) Do students need to be US Citizens to participate in the project?

No, students do not need to be US citizens to participate. The project will utilize open-source datasets, ensuring no access restrictions related to citizenship or security clearance.

3) What backgrounds or prerequisites would you prefer that students have?

Preferred skills and knowledge include:

- **Programming:** Proficiency in Python, JavaScript, or R for data processing and visualization.
- **Geospatial Tools:** Experience with GIS platforms such as QGIS, OpenStreetMap, or Google Maps APIs.
- **Database Management:** Familiarity with SQL or NoSQL databases for handling and organizing large datasets.
- **User Interface Design:** Skills in developing accessible, user-friendly applications.
- **Teamwork & Communication:** Ability to collaborate effectively and engage with external stakeholders.
- **Curiosity and Problem-Solving:** A proactive mindset to explore innovative solutions.

4) How many hours per week do you expect from each student on your team?

Students should dedicate **8–10 hours per week** to the project, including team collaboration, individual contributions, and attending regular meetings.

5) Please include here any links or data or websites that might help students to better understand the overall nature of the focus of the project.

- **WomanStats Database:** <https://www.womanstats.org>
- **OpenStreetMap:** <https://www.openstreetmap.org>
- **HHS Open Data:** <https://healthdata.gov>
- **UN Women Resources:** <https://www.unwomen.org>
- **Geospatial Data Tutorials:** <https://learnosm.org>

6) In one paragraph, please describe the expectations you have of your student team with respect to the project, deliverables, meetings, engagement, etc.

We expect the student team to demonstrate a high level of initiative, collaboration, and professionalism. Team members should actively engage in biweekly meetings to discuss progress, challenges, and next steps. Deliverables will include a functional geospatial platform, documentation, and a final presentation.

detailing the solution. While the team is encouraged to work autonomously, regular communication and updates are essential to ensure alignment with project goals. Students should take ownership of their respective tasks while supporting one another to overcome obstacles and achieve a cohesive outcome.

7) Please include anything else here that will help me, you, and students in creating appropriate and successful teams.

This project offers students a chance to address real-world humanitarian challenges while developing practical skills in geospatial analysis, software development, and data visualization. Working with a client like Q2 Impact and the WPS team at NORTHCOM provides unique exposure to both public and private sector collaboration. Students will gain hands-on experience with tools and datasets that directly impact emergency response and policy-making, making this an excellent opportunity for those passionate about applying technology for social good.